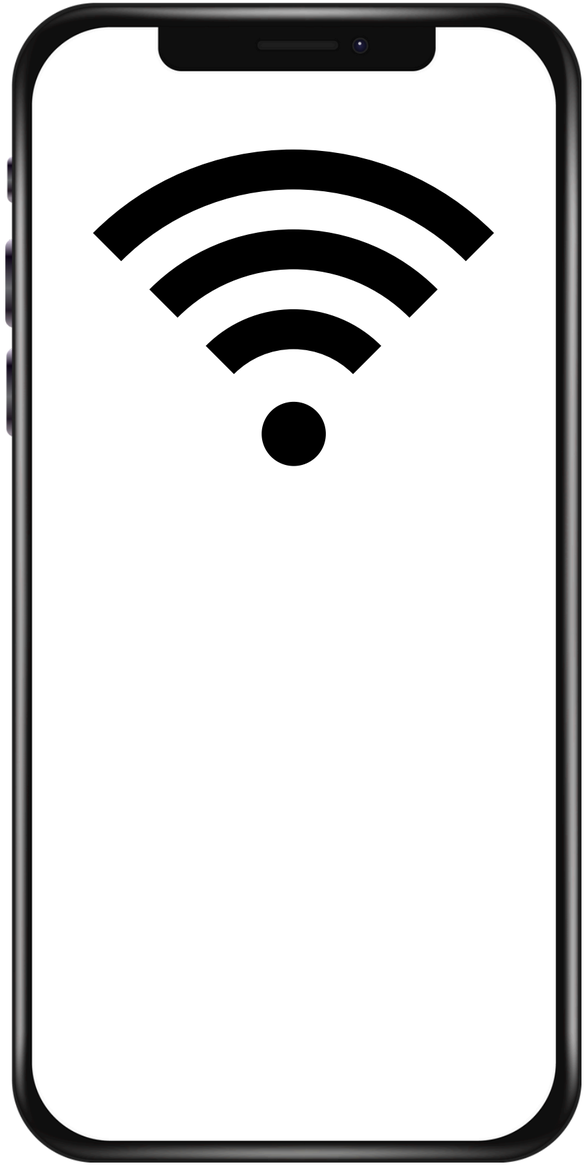
COMPUTER NETWORKS

CSE 321

What do you say at first while visiting any relative’s house?



“What is the Wi-Fi password”

Obtaining
IP address

Motivation

When you enter into a network, how will you know your IP address, subnet mask, gateway etc?

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Commonly used for servers, routers etc.

But is it appropriate in dynamic scenario?

DHCP

Dynamic Host Configuration Protocol

Automatically assigns IP addresses, subnet masks, default gateways etc. (IP will be valid up to a certain duration called lease time)

DHCP Components

- **Client –** User who wants to connect to the network
- **DHCP Server –** Manages configuration
- **DHCP Relay Agent –** If server and client is on a different subnet, relay agent is needed to relay the messages from the client

DHCP Basic Operation

DHCP Basic Operation

DHCP Basic Operation

Discover

Offer

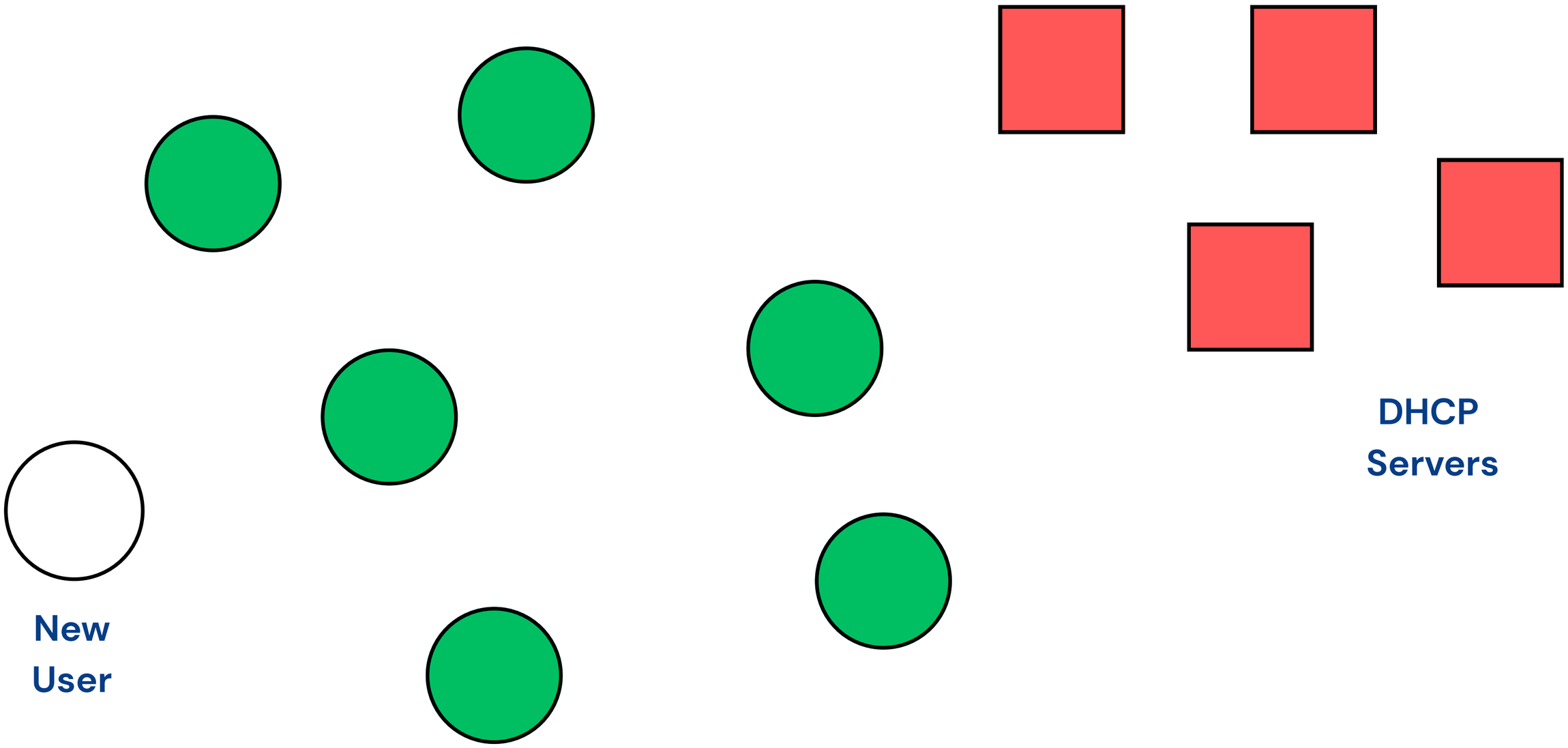
Request

Acknowledgement

Discover

The new user will search for servers

Discover (without Agent)



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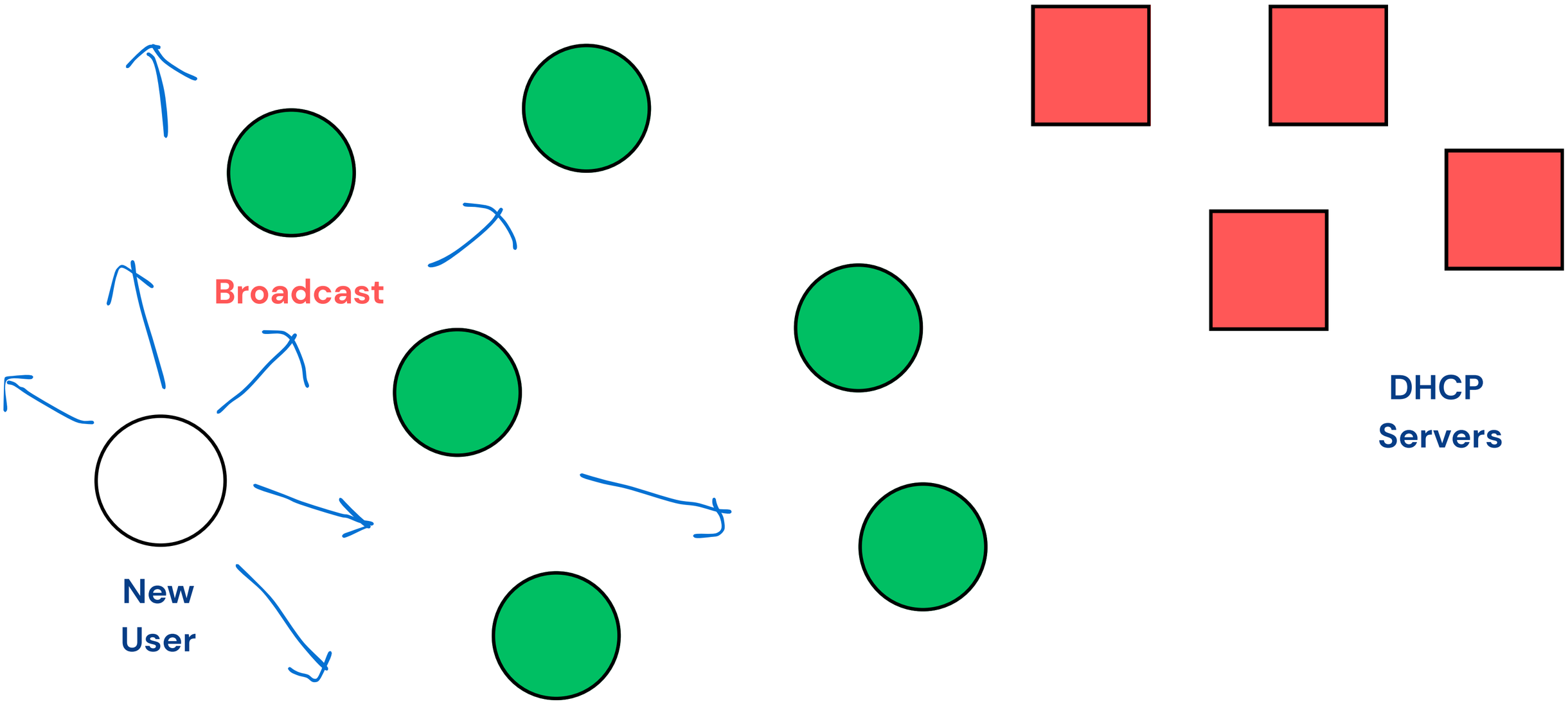

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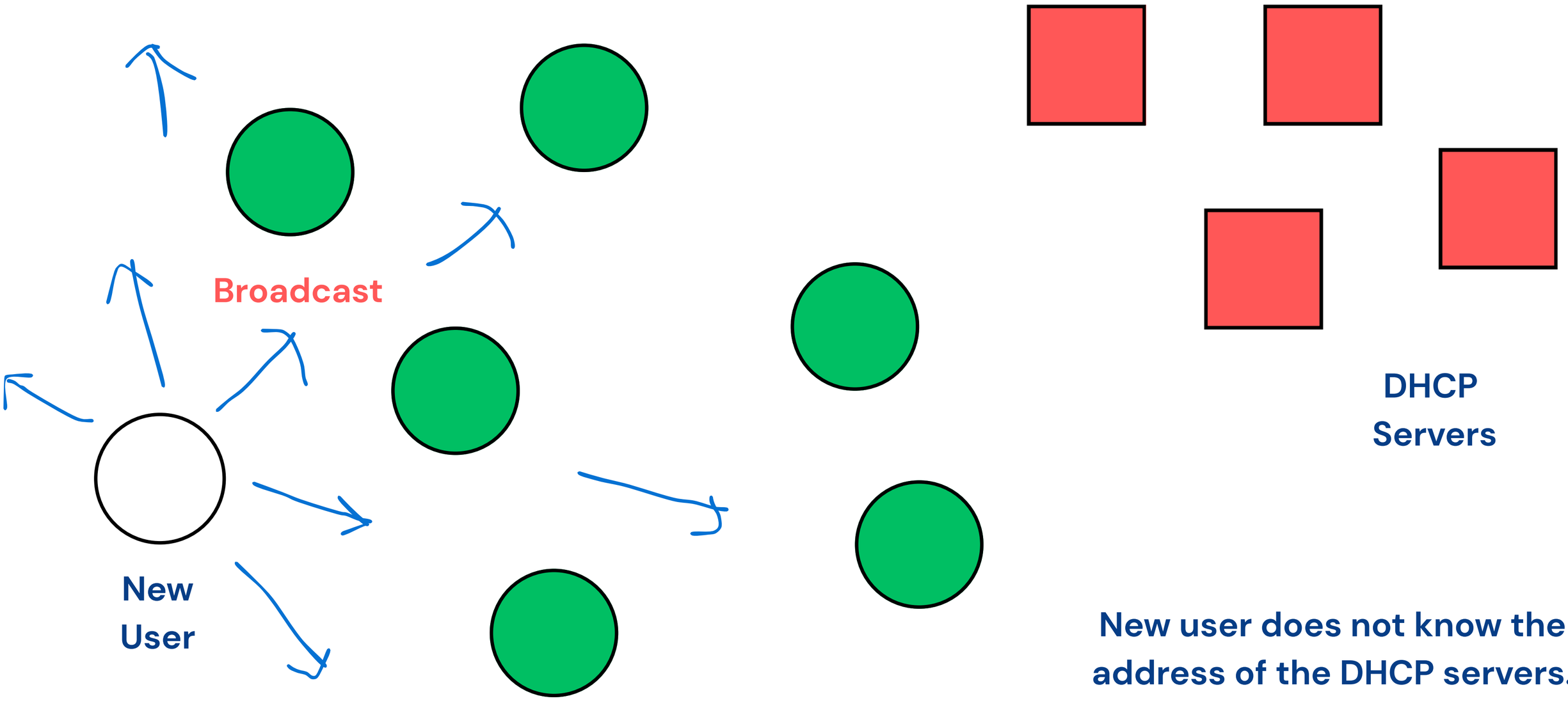
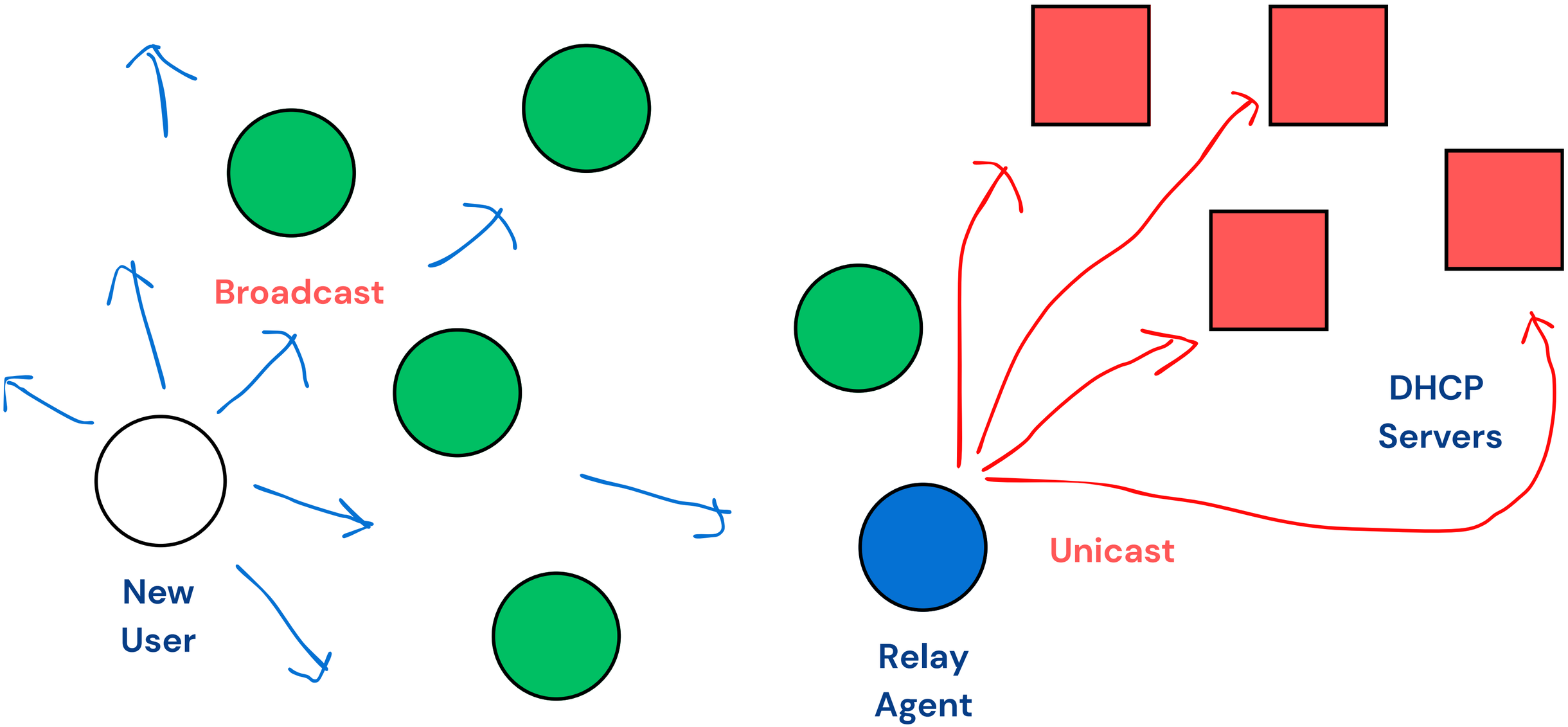


New user does not know the address of the DHCP servers. That is why, it broadcasts Discover message

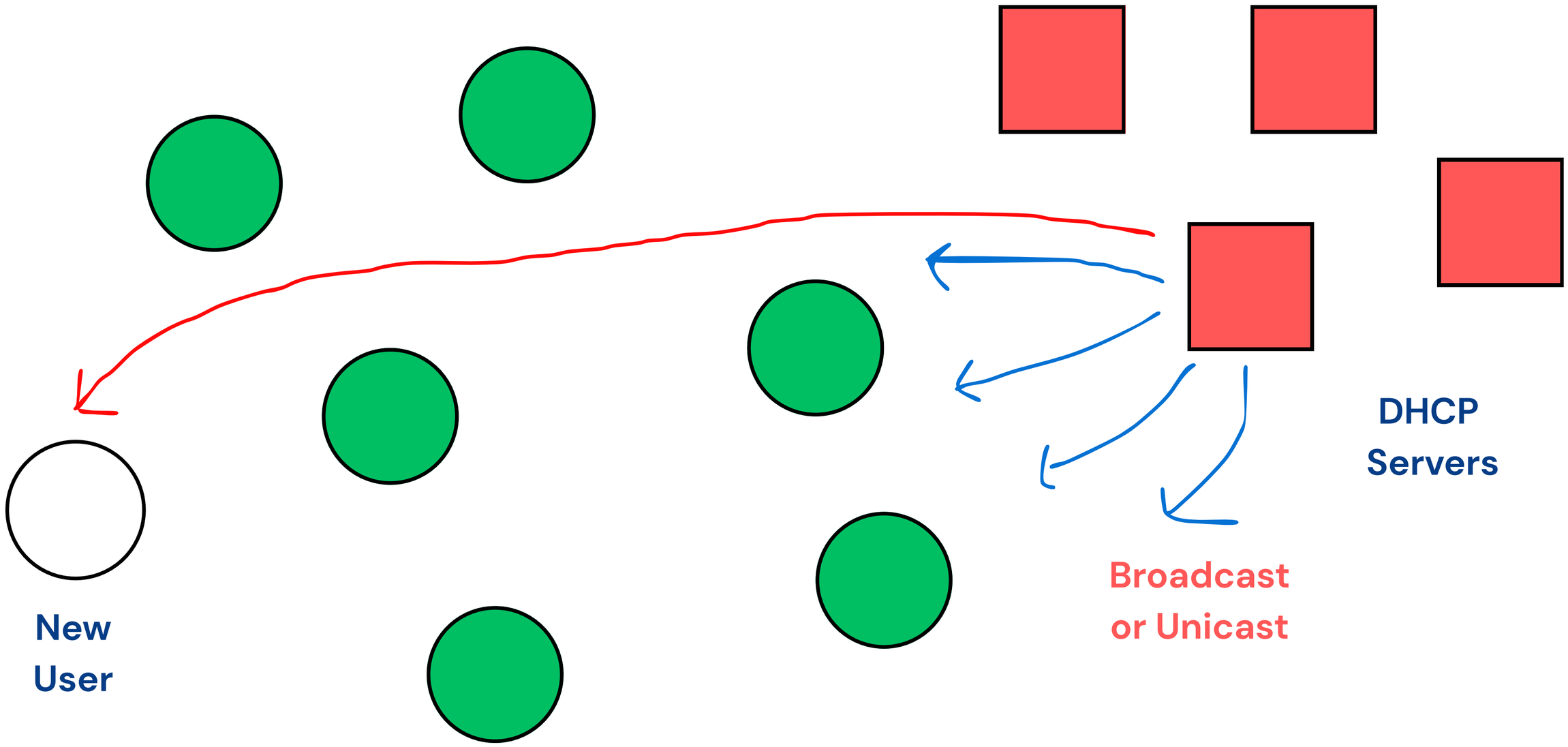
Discover (with Agent)



Offer

DHCP servers (all available servers) will offer IP and configurations to the user

Offer (without Agent)



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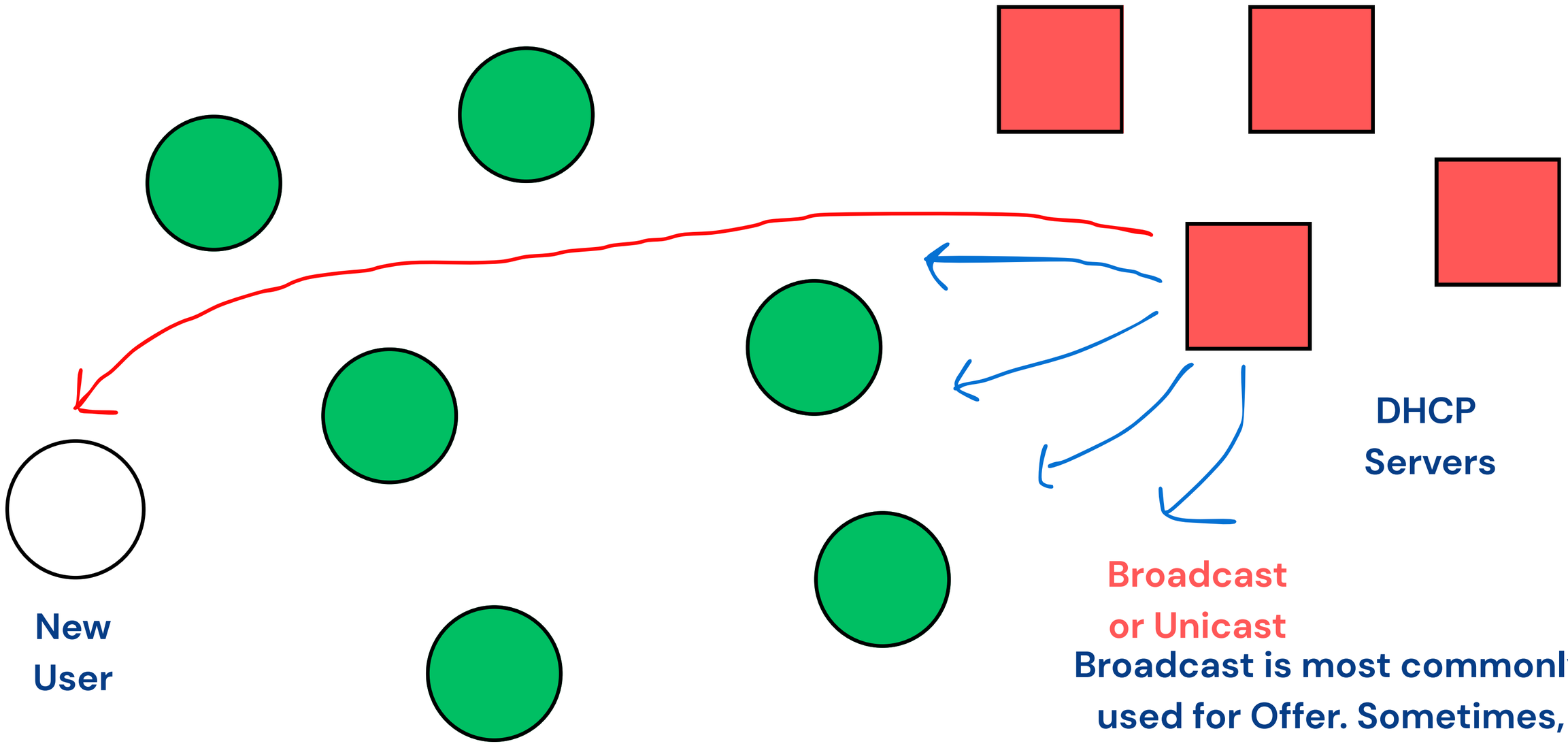
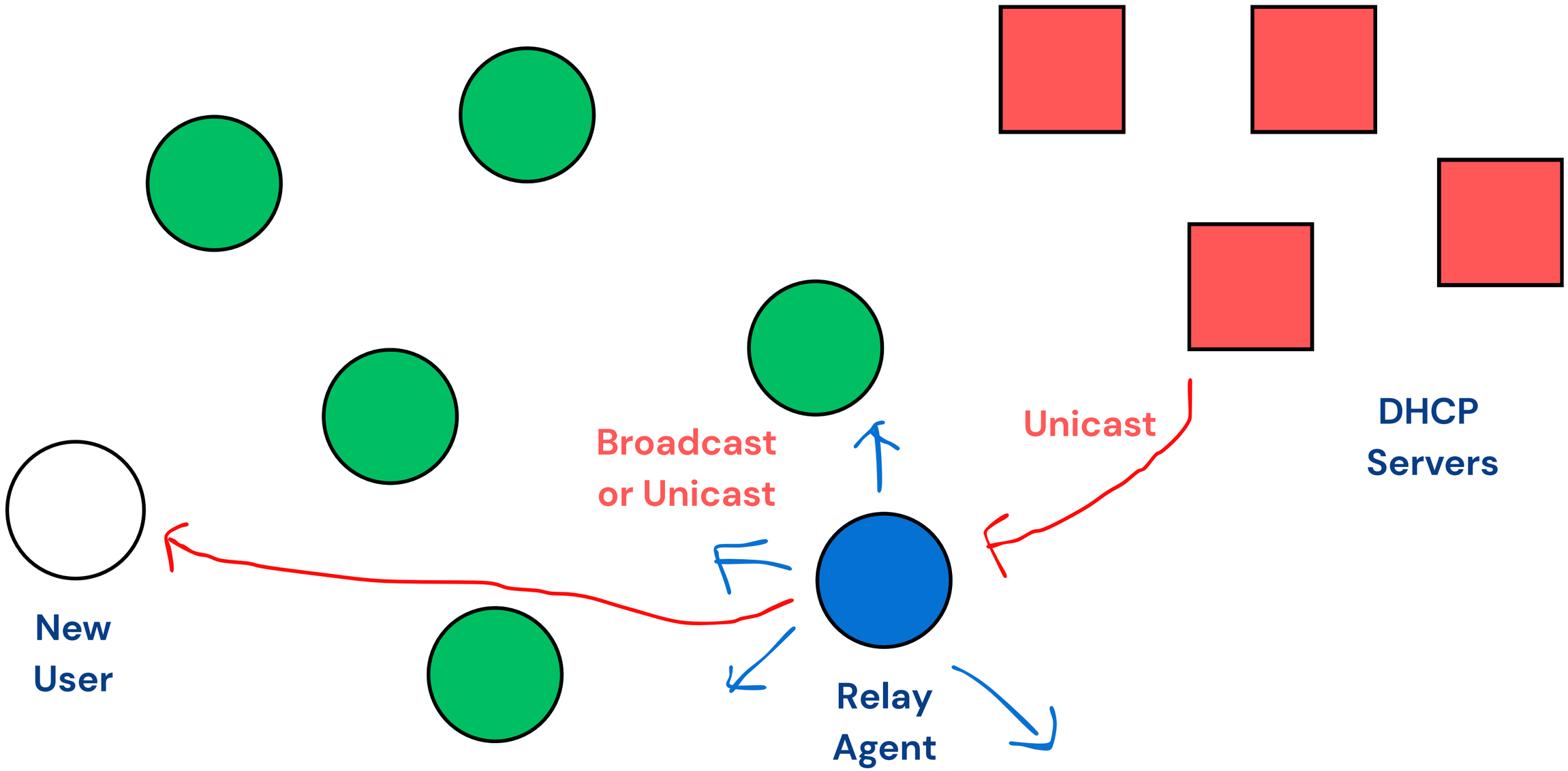


Broadcast is most commonly used for Offer. Sometimes, unicast is done using MAC address.

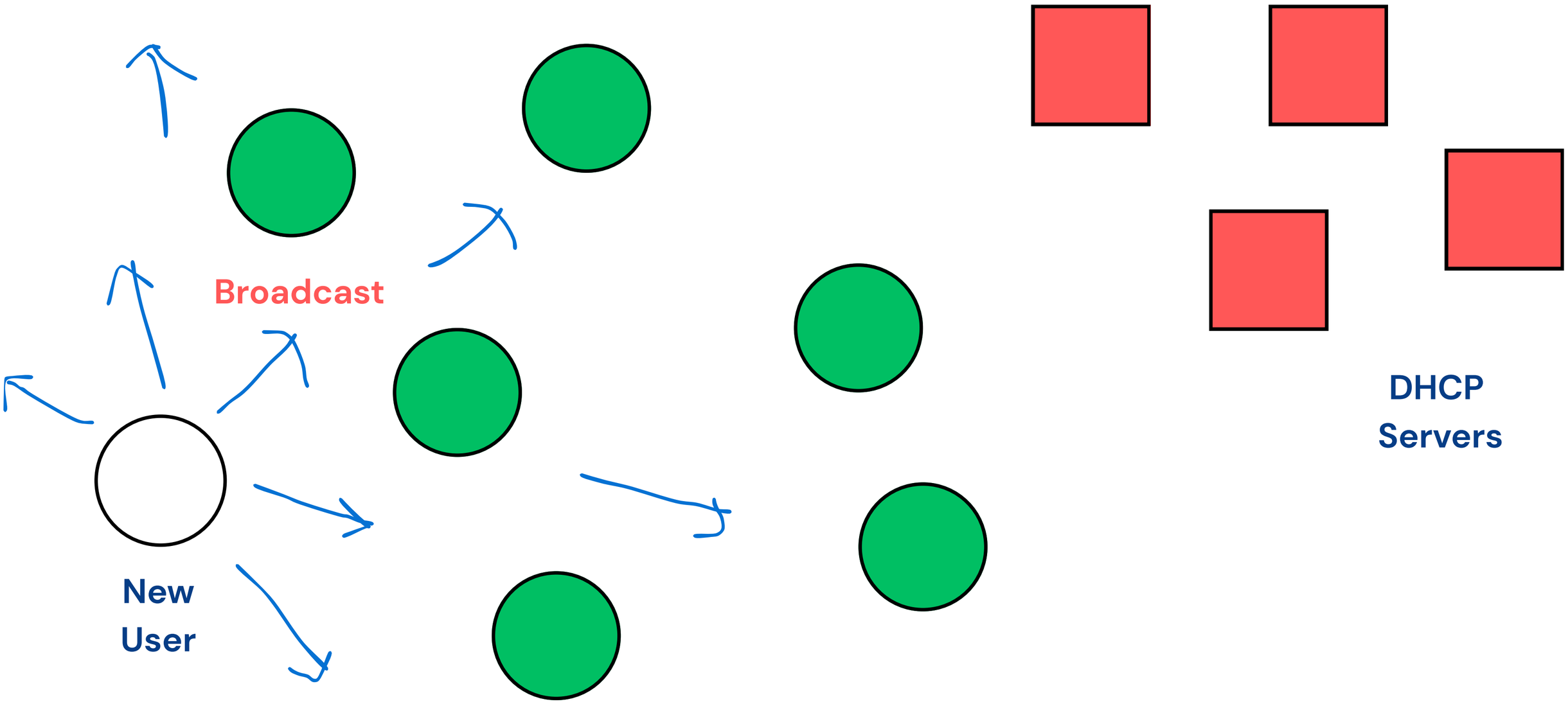
Offer (with Agent)



Request

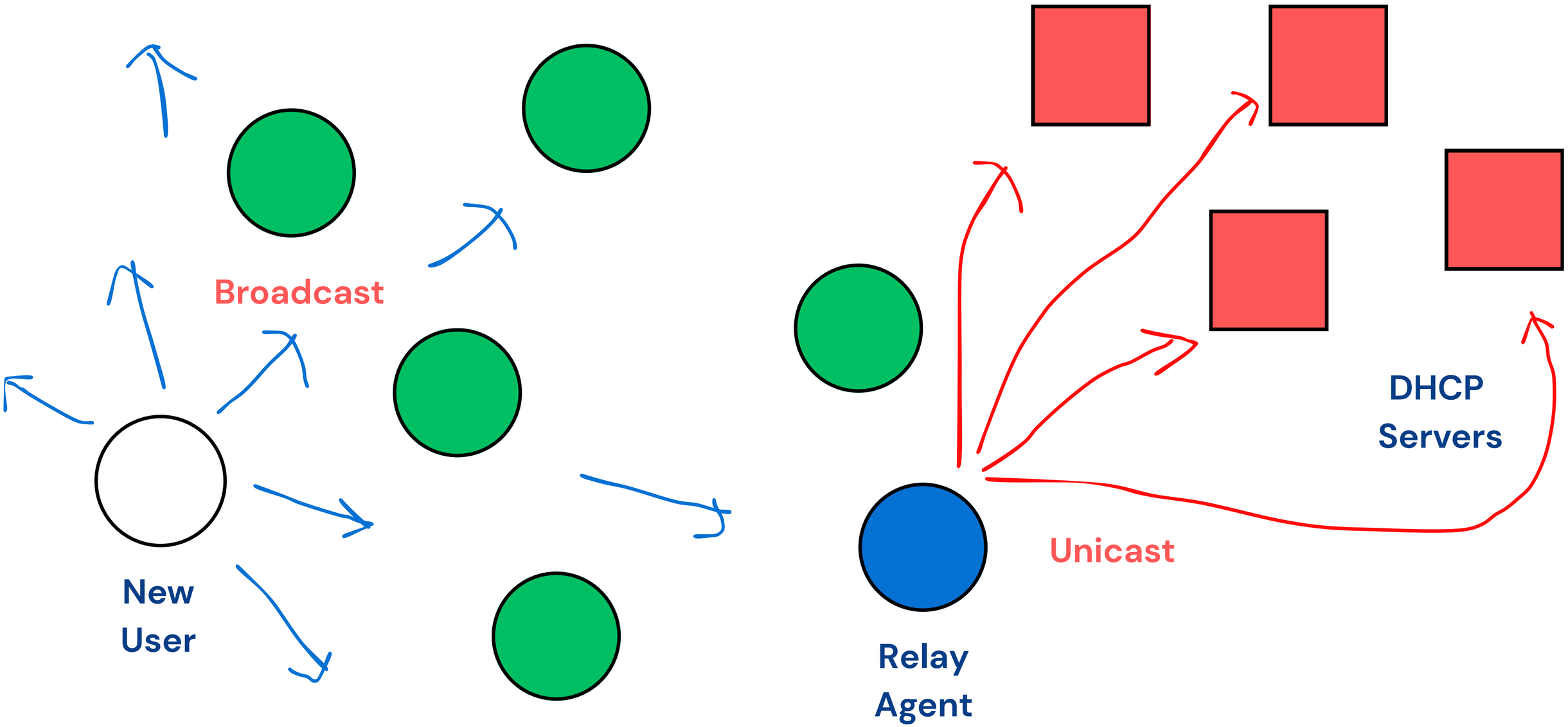
User will choose an offer and informs the server

Request (without Agent)



Similarly as Discover

Request (with Agent)



Similarly as Discover

Request

User will choose an offer and informs the server

Why is this message sent to other DHCP servers as well?

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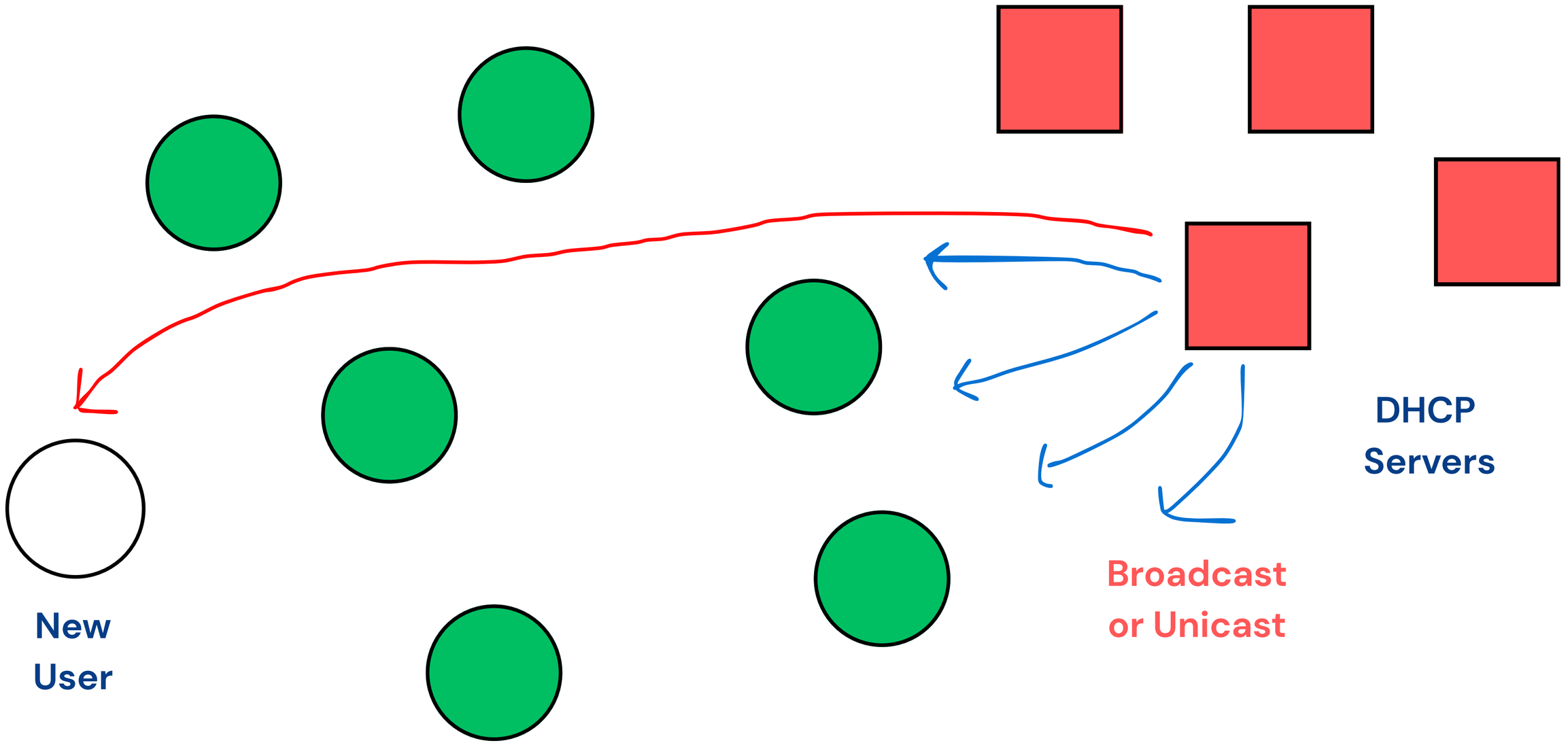
Why is this message sent to other DHCP servers as well?

To let other servers know which server is chosen by the user. They will withdraw their offers.

Acknowledgement

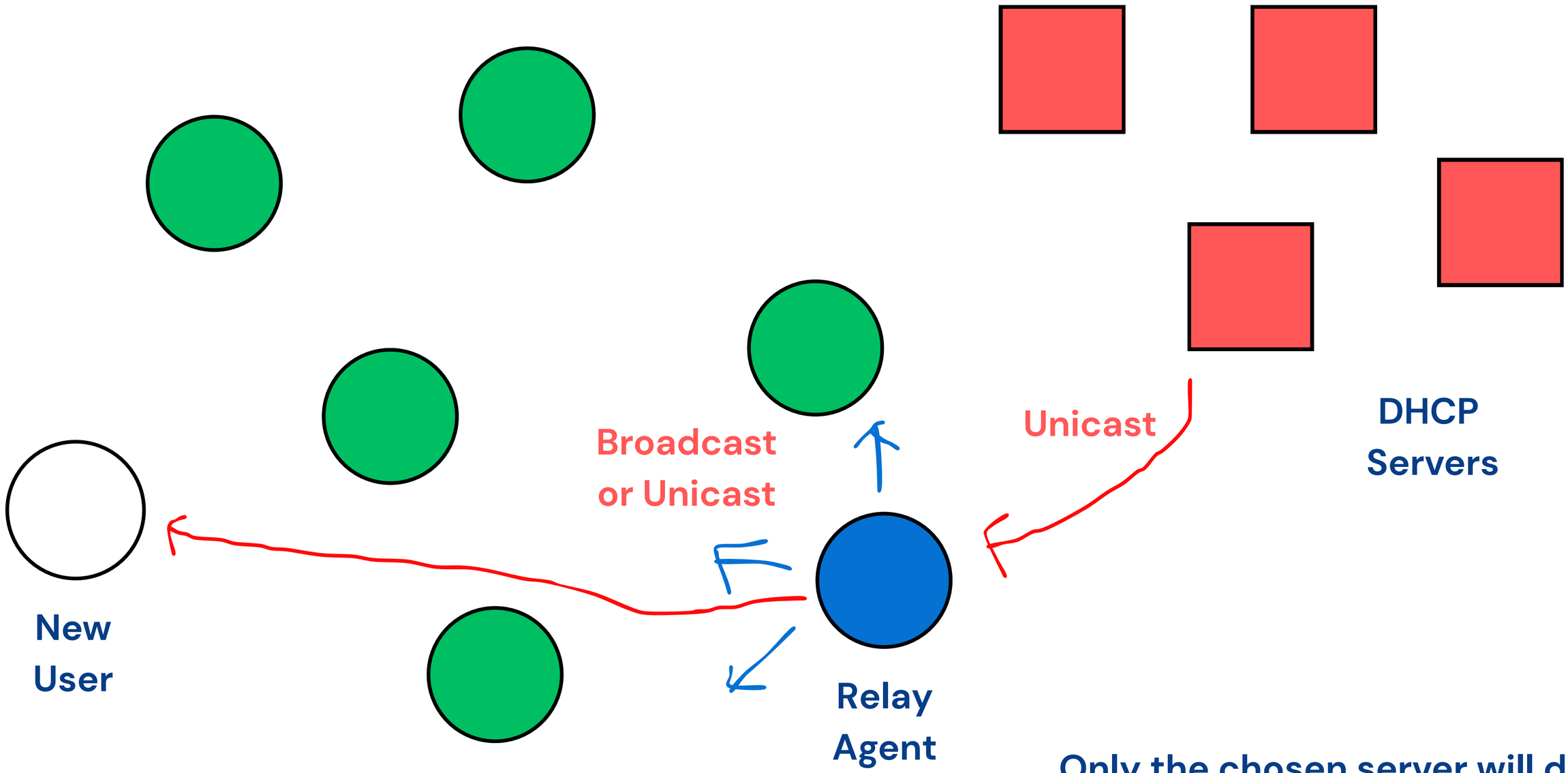
The server confirms the assignment and logs the lease in its database.

Acknowledgement (without Agent)



Only the chosen server will do this

Acknowledgement (with Agent)



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DHCP Renewal

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After T1 time (generally 50% of the lease time)

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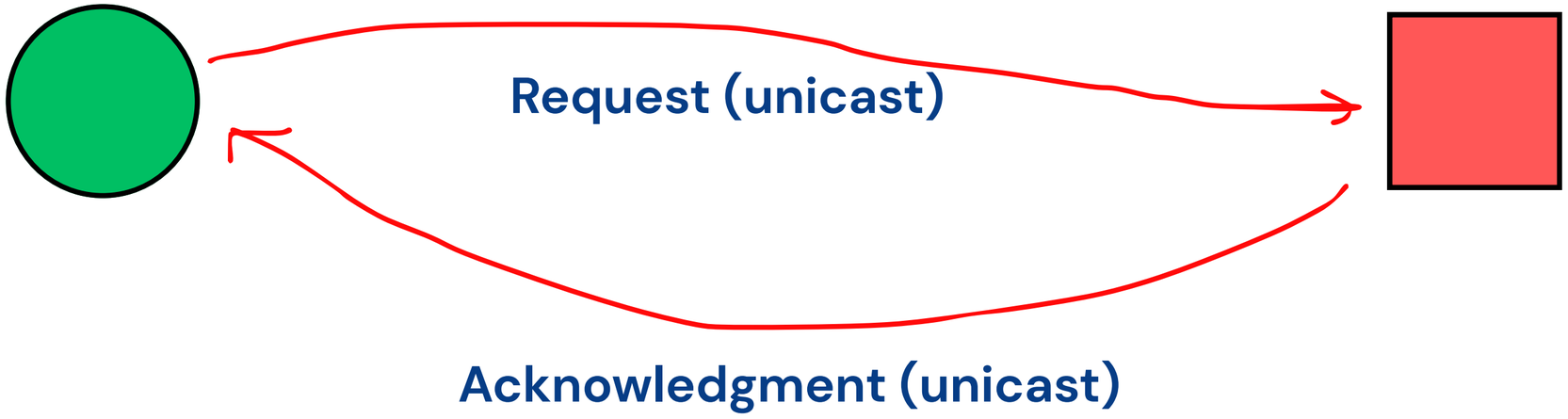

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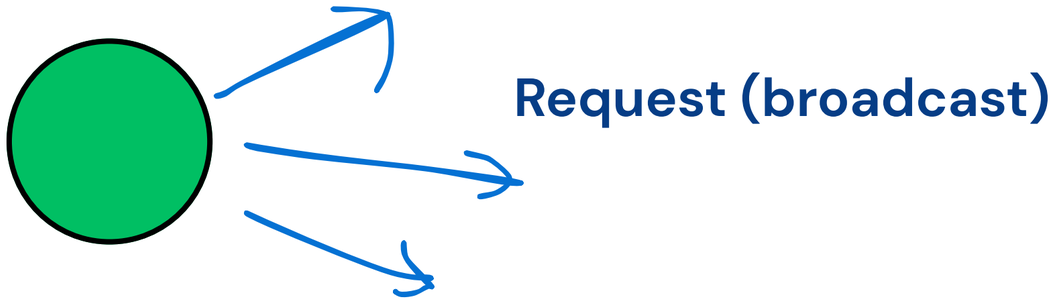
After T1 time (generally 50% of the lease time) (if failed)

After T2 time (generally 87.5% of the lease time)



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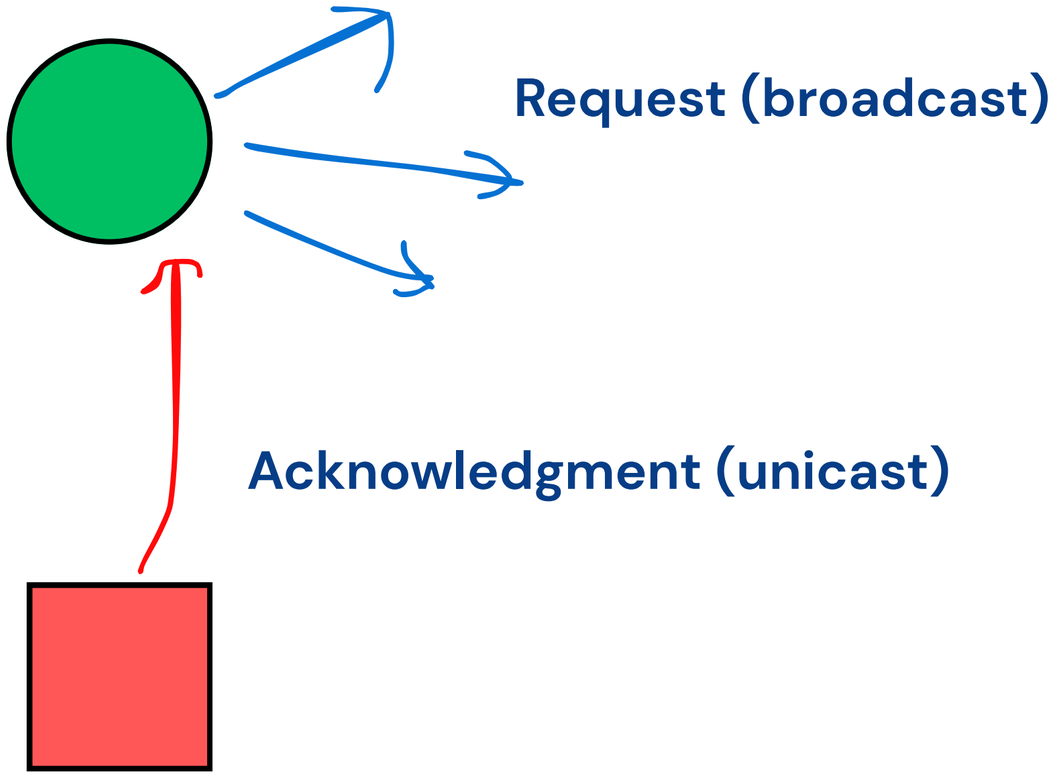

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After T1 time (generally 50% of the lease time) (if failed)

After T2 time (generally 87.5% of the lease time) (if failed)

After 100% of the lease time

DHCP SERVER
DEVICE CONNECTING TO THE NETWORK
IP ADDRESS
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